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SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02 GROUP 04

TECHNICAL REPORT

TOPIC: DATA ARCHITECTURE IN DATA ENGINEERING

LECTURER: DR. ARYATI BINTI BAKRI

GROUP MEMBER:

STUDENT NAME	MATRIC NO
SABRINA HENG WEI QI	A23CS0265
GUI KAH SIN	A23CS0080

DATA ARCHITECTURE IN DATA ENGINEERING

Executive Summary

A data architecture can be considered a crucial foundation for today's companies, which rely on big data for making decisions. The paper analyses the development of data architecture from traditional data warehouses to new-generation, highly agile and elastic solutions like Data Lakehouse. Breaking down all key layers of the data architecture, namely ingestion, storage, processing and serving, we examine the mechanisms by which companies handle the vast amounts of different big data.

While the modernization of such systems allows for reaping many advantages, it is not devoid of difficulties. One should mention high infrastructure expenses and the daily management of big data, among other issues.

To overcome them, our analysis proposes focusing on scalable solutions based on the cloud architecture, creating strict data governance policies in the early stages of the process and using modern processing methods. However, even the latest technologies make up only one side of the issue since data architecture should also match company's goals and be accompanied by professional engineers

1. Introduction

In the world of digital business today, data is often referred to as the most valuable asset that an enterprise can possess. But on its own, raw data has little value, and for it to have business value, it must be collected, structured and processed. The data engineering domain is the one whose base is data architecture. Data architecture is the comprehensive framework that describes how data is collected, stored, processed, shared and used throughout the organization.

The success of enterprise systems depends on data architecture almost entirely. A lack of a coherent architectural vision leads organisations into data silos, inconsistent formats, slow retrieval and poor decision-making with operational inefficiencies. A robustly architected data flow from operational systems to analytical systems enables advanced capabilities such as predictive modeling and real-time reporting.

This report aims to simplify the data architecture aspect within data engineering. This has something to do with the basic building blocks of data infrastructure, data pipelines and how they are applied in enterprise systems. This paper studies integration of different theoretical frameworks and practical case studies that will help you understand how modern data architecture drives business success.

2. Background / Literature Review

In order to understand the current state of data architecture, first must take a look at how quickly the field has progressed due to the ever-changing needs of business. Initially, the main issue was handling highly structured data, such as financial transactions and inventory. Ralph Kimball and Bill Inmon provided the initial methodologies behind the creation of traditional data warehouses (Kimball & Ross, 2013). They were highly structured and efficient for reporting, but were very rigid since all elements had to be structured in accordance with rows and columns.

According to Doug Laney (2001), the rise of the era of Big Data increased both the number and complexity of data sources. The introduction of smartphones, social media and smart devices introduced an immense amount of unstructured data such as photos, text logs and other sensor readings. It made traditional data warehouses too expensive and inflexible to accommodate the increasing amount of information. This led to a change of paradigm, the industry developed Data Lake architecture based on big data technologies, such as Apache Hadoop. According to Zikopoulos et al. (2011), data lakes allow storing vast amounts of raw data in its native format using low costs. At the same time, another issue became prominent, where data lakes often become a mess because the data lacked systemization, creating “data swamps”.

Thus, the current paradigm can be considered as the Data Lakehouse which combines the most beneficial elements from the two previous ones. Lakehouse allow storing large amounts of data in their natural format, like data lakes, but at the same time restore structure and performance of the warehouse.

Along with this shift in architectures, another development that significantly changes the nature of work with data is the transition to cloud providers like AWS, Google Cloud and Microsoft Azure. Instead of working with physical servers and doing overnight batch processing, data engineers now use serverless architectures of the clouds capable of processing real-time streaming data (Kleppmann, 2017).

Although the described developments can be considered highly advanced, existing literature points out the major issues that have yet to be addressed. Namely, the problem of automated data governance becomes particularly relevant nowadays, as cloud computing increases the distribution and complexity of enterprise data architecture.

3. Analysis and Discussion

Data architecture in data engineering is a guidance of how data is collected, stored, processed, and delivered for analytical and operational use. It is a blueprint that makes sure data flows efficiently across systems while ensuring quality, scalability, and accessibility. In modern data engineering environments, a layered approach to data architecture is often used to integrate different data processes into cohesive layers such as data sources, ingestion, processing, and data serving.

3.1 How the Technology Works

The functionality of data architecture can be analyzed through its structured layers. Data created from multiple sources such as transactional databases, enterprise systems, APIs, and IoT devices (Reis, J., & Housley, M., 2022). There are several data sources that produce a large amount of structured and unstructured data that must be integrated and processed efficiently.

The ingestion component collects data through batch or real-time streaming methods, in order to collect reliable data into the system. According to Reis and Housley (2022), this stage is essential for capturing data before it moved into the storage layer. Then, the storage component manages data using scalable platforms such as data lakes, data warehouses, and data lakehouses, based on the type of data used.

Furthermore, the processing component transforms raw data into relevant and usable formats through ETL (Extract, Transform, Load) or ELT (Extract, Load, Transform) techniques. It is responsible for maintaining data quality, consistency, and integration across systems. Lastly, the serving component sends processed data to the end users via dashboards, reports, or machine learning models to improve decision-making. This layered structure makes sure data flow efficiently and supports scalability and flexibility in modern data environments.

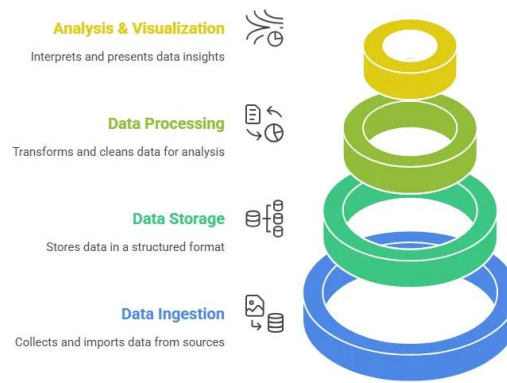


Figure 1: Layer of Data Architecture

3.2 Case Examples or Applications

In real world, organizations such as Netflix and Amazon utilize well-designed data architectures to manage large-scale data operations. Netflix uses a massive amount of user interaction data to customize personalized recommendations, demonstrating the importance of real-time data processing and scalable architecture. Similarly, Amazon uses cloud-based data architecture to handle customer data, optimize operations, and support advanced analytics. These examples highlight how data architecture enables organizations to deal with big data characteristics (3V: volume, velocity, and variety) while helping them to enhance business efficiency and decision-making capabilities (Avci et al. 2020).

Beyond the entertainments, healthcare systems are using Data Mesh architectures. In the systems, departments like radiology own their specific data products, ensuring that domain experts maintain high data quality while allowing the whole hospital system to work together (Werner et al., 2026). Thus, it shows the importance of aligning architecture with the data engineering lifecycle to ensure the seamless and effective data flow.

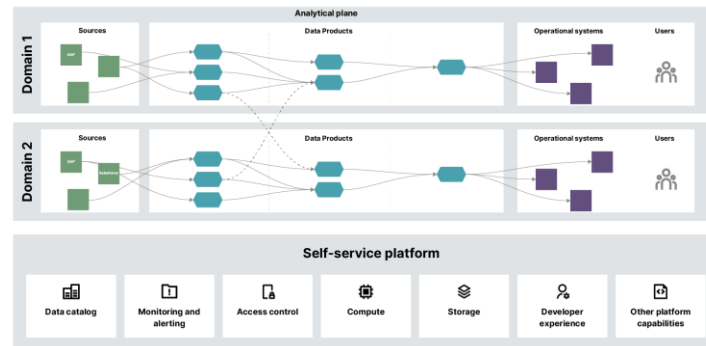


Figure 2: Data Mesh Architecture

3.3 Advantages and Disadvantages

Data architecture offers several advantages. It enhances scalability, allowing organizations to manage growing data volumes efficiently. It also supports flexibility by accommodating different data types and integrating multiple data sources. Besides, a well-structured architecture will improve data accessibility and enables faster and more accurate decision-making (Inmon et al., 2019).

However, there are also some disadvantages. The design and implementation of data architecture can be complex and require technical knowledge. Moreover, the cost of infrastructure can be extremely high especially in cloud-based environments. As mentioned by Abughazala (2024), the governance of data, including issues like data quality, security, and compliance as the architecture expands, are a severe challenge.

3.4 Challenges and Future Trends

One of the biggest challenges faced is data silos because they limit effective data integration across departments. It is not easy to maintain the quality of data within different sources. Besides, organizations need to take into consideration data security and privacy concerns that have been raised within regulated industries (Abughazala, 2024).

Future trends can play an important role in influencing the evolution of data architecture. The data lakehouse architecture combines strengths from both data warehouse and data lake, allowing high flexibility and performance. Data mesh and data fabric are some new ideas that encourage decentralized data ownership and improved integration. However, cloud-based architecture and real-time data processing technologies cannot be overlooked too (Plazotta & Klettke, 2024).

4. Recommendations

Based on the analysis, organizations should acquire a strategic and appropriate approach when designing and implementing data architecture in data engineering. First, they should use a layer architecture approach to ensure scalability and modularity. It is an excellent idea to adopt because it allows different components of the system to operate independently while maintaining efficient data flow (Avci et al., 2020). Furthermore, organizations should also consider using cloud-based platforms since it provides scalable storage and processing capabilities that enables them to handle large amounts of data without handling infrastructure complexity (Plazatto & Klettke, 2024). They should also develop appropriate data governance strategies, which includes setting up policies regarding data quality and management. Additionally, organizations should invest in modern data processing technologies, especially the distributed computing systems to enable them to perform batch and real-time analytics. It will improve their decision-making and overall performance. They should also focus on developing and training skilled data engineers who can work with complex data architectures and the evolving technologies (Inmon et al., 2019).

5. Conclusion

Overall, data architecture is an important component of data engineering, as it provides a structured blueprint for managing the flow of data across an organization. By applying the layered architecture approach, organizations can effectively collect, store, process, and deliver data for analytical and operational purposes (Inmon et al., 2019).

While data architecture offers significant benefits such as scalability, flexibility, and better decision-making, it also has challenges related to complexity, cost, and data governance. Careful planning, appropriate technology adoption, and skilled expertise can greatly solve those challenges.

Data lakehouse architecture, data mesh, and cloud-based solutions are all new data platforms that are expected to improve data architecture even more in the future. These improvements will help businesses deal with complicated data environments and get the greatest value from their data.

References

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Appendix / Logbook

Github Link: https://github.com/SabrinaHeng/SPDE_Technical_Report/tree/main

AI Tools and Usage

Date	AI Tool Used	Purpose of Use	Prompt Used	Verification Method
25 Mar	ChatGPT	Helped structure report outline	Help me create a technical report outline for Data Architecture in Data Engineering	Verified with assignment template provided by lecturer
26 Mar	ChatGPT	Explained basic concept of data architecture	Explain data architecture in data engineering in a simple and clear way	Cross-checked with academic sources and lecture materials
27 Mar	Gemini	Explained data mesh and real-world relevance	Explain data mesh in easy understand way and how does it relate to healthcare system	Compared with online articles and journals
28 Mar	Gemini	Generated research keywords	What is the keyword for Data Architecture in Data Engineering	Used keywords to search in Google Scholar and verified relevance
29 Mar	Gemini	Helped understand data architecture concept	Please explain the data architecture concept to me	Cross-checked with multiple sources and lecture notes
5 Apr	ChatGPT	Improved writing clarity and grammar	Check grammar and improve this conclusion paragraph	Manually reviewed to ensure meaning remains accurate
10 Apr	Turnitin	Plagiarism checking	N/A	Confirmed similarity score is within acceptable limit

Appendix Screenshots

Explain data architecture in data engineering clearly with examples

🌱 What is Data Architecture in Data Engineering?

Data architecture in data engineering is the overall design (blueprint) that defines how data is:

- collected
- stored
- processed
- managed
- delivered

across an organization.

👉 In simple terms:

It is the plan that ensures the right data flows to the right place, at the right time, in the right format.

Explain data mesh in easy understand way and how does it relate to healthcare system



To understand **Data Mesh** easily, think of it as a shift from a "Centralized Kitchen" to a "Food Court."

1. Easy Explanation: The Food Court Analogy

- **Traditional Model (Centralized Kitchen):** All the ingredients (data) are sent to one central kitchen (IT team). If someone wants a burger, a salad, or sushi, they all have to wait in one long line for the same group of chefs to prepare it. This creates a **bottleneck** because the chefs might not know the "secret recipe" for every specific dish.
- **Data Mesh (The Food Court):** Instead of one kitchen, each specialized shop (e.g., Radiology, Pharmacy) has its own kitchen, chefs, and ingredients. They are the experts in

Detailed report

20/03/2026 - 12/04/2026



Total: 21:46:56 Billable: 21:46:56 Amount: 0,00 USD

Date	Description	Duration	User
10/04/2026	Finalize Report Topic 8 - Final Editing	00:29:24 23:27:00 - 23:56:24	Sabrina Heng 0,00 USD
10/04/2026	Done Logbook Topic 8 - Logbook	00:38:23 09:00:00 - 09:38:23	Sabrina Heng 0,00 USD
09/04/2026	Changes in Analysis & Discussion Topic 8 - Write Analysis & Discussion	00:59:00 22:13:00 - 23:12:00	GUIKAHSIN 0,00 USD
09/04/2026	Finalize References Topic 8 - Reference	00:12:36 12:00:00 - 12:12:36	Sabrina Heng 0,00 USD
07/04/2026	Insert and Check References Topic 8 - Reference	00:22:19 17:00:00 - 17:22:19	Sabrina Heng 0,00 USD
06/04/2026	Write Executive Summary Topic 8 - Write Executive Summary	00:33:14 21:00:00 - 21:33:14	Sabrina Heng 0,00 USD
05/04/2026	Write Conclusion Topic 8 - Write Conclusion	01:17:00 20:38:00 - 21:55:00	GUIKAHSIN 0,00 USD
05/04/2026	Add Reference Used Topic 8 - Reference	00:12:00 16:31:00 - 16:43:00	GUIKAHSIN 0,00 USD
05/04/2026	Write Recommendations Topic 8 - Write Recommendations	01:13:00 15:16:00 - 16:29:00	GUIKAHSIN 0,00 USD
03/04/2026	Add Reference Used Topic 8 - Reference	00:15:00 16:48:00 - 17:03:00	GUIKAHSIN 0,00 USD
03/04/2026	Write Analysis & Discussion (Part 2) Topic 8 - Write Analysis & Discussion	02:16:14 14:30:00 - 16:46:14	GUIKAHSIN 0,00 USD
01/04/2026	Add Reference Used Topic 8 - Reference	00:18:00 17:55:00 - 18:13:00	GUIKAHSIN 0,00 USD
01/04/2026	Write Analysis & Discussion (Part 1) Topic 8 - Write Analysis & Discussion	03:06:00 14:47:00 - 17:53:00	GUIKAHSIN 0,00 USD
31/03/2026	Write Literature Review (Part 2) Topic 8 - Write Literature Review	00:21:43 09:00:00 - 09:21:43	Sabrina Heng 0,00 USD
30/03/2026	Write Literature Review (Part 1) Topic 8 - Write Literature Review	00:36:58 09:00:00 - 09:36:58	Sabrina Heng 0,00 USD

29/03/2026	Write Introduction Topic 8 - Write Introduction	00:42:21 18:00:00 - 18:42:21	Sabrina Heng 0,00 USD
28/03/2026	Research on Background Data Architecture Topic 8 - Research for the topic	01:12:03 22:00:00 - 23:12:03	Sabrina Heng 0,00 USD
28/03/2026	Research on Data Architecture Topic 8 - Research for the topic	03:05:00 10:02:00 - 13:07:00	GUIKAHSIN 0,00 USD
28/03/2026	Explore Searching Tools Topic 8 - Explore searching materials	00:59:00 09:03:00 - 10:02:00	GUIKAHSIN 0,00 USD
27/03/2026	Explore searching materials Topic 8 - Explore searching materials	00:26:36 14:00:00 - 14:26:36	Sabrina Heng 0,00 USD
27/03/2026	Research on Data Architecture Topic 8 - Research for the topic	01:38:00 10:26:00 - 12:04:00	Sabrina Heng 0,00 USD
26/03/2026	Draft report outline using template Topic 8 - Drafted report outline	00:15:41 15:00:00 - 15:15:41	Sabrina Heng 0,00 USD
25/03/2026	Discussion for Distribute Task Topic 8 - Distribute Tasks	00:37:24 09:00:00 - 09:37:24	Sabrina Heng 0,00 USD